Report on

Net Metering Intervention and challenges in Dhapsung community solar micro grid

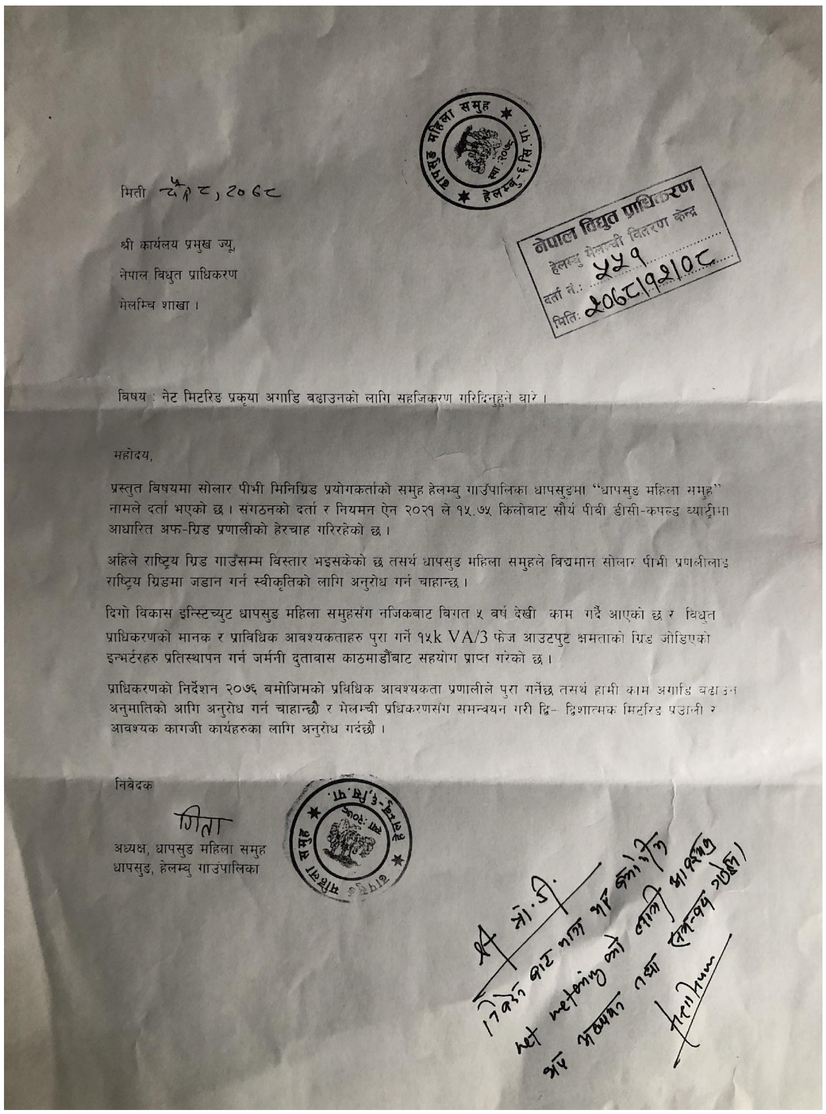
Figure SEQ Figure * ARABIC 1: A 16 KW Dhapsung community solar micro grid

In 2016, the Digo Bikas Institute installed a 16kW solar microgrid in Dhapsung village, a rural part of Sindhupalchowk district in Nepal with partner organizations Grid Alternatives and Gham Power. Prior to the installation, the village had a 3kW Peltric set for basic lighting, which was destroyed in the 2015 earthquake. The 16kW solar microgrid is the first of its kind to be owned and operated by a women's group in Nepal, the Dhapsung Women Group. The group was responsible for setting the tariff rates for the electricity generated with input from the users. The solar microgrid also had social empowerment benefits for the community. The income generated from the microgrid also increased the community's access to microloans provided by the Women Group at a 10% interest rate for small businesses or emergencies.

Later, the national transmission line was connected to Dhapsung, and the solar microgrid has been inactive since then. The transformation of the people of Dhapsung from active consumers of the solar microgrid to passive consumers of the national utility has resulted in the loss of socioeconomic benefits from the microgrid. To address this issue, the plan to introduce a net metering facility for the Dhapsung solar microgrid had been endorsed with the agreement of the Women Group and the user group. Nepal introduced a net metering policy and directive in 2018, which allows consumers to offset the amount of solar energy they export to the grid against the amount of grid energy they import.

Digo Bikas Institute coordinated with the NEA Melamchi to connect the microgrid to the national grid through Dhapsung Women Group. For this, the women group had submitted the letter to NEA

Melamchi. On 3rd January 2022, DBI and Dhapsung women group representative had a meeting with Santosh Prasad Pant, Chief of NEA at Melamchi, and shared the solar microgrid work in Dhapsung. He requested the detailed specification of the Dhapsung solar microgrid, along with its last functional status and year since it was in operation. He mentioned after receiving the details, he will discuss further with the power purchase agreement team on this matter. There was not any local institution in Dhapsung and Nepal Electricity Authority (NEA), Melamchi distribution suggested for legal registration of the Dhapsung Women group with whom the power purchase agreement could be signed. As per the suggestion from NEA and following the net metering policy documenys, Dhapsung women group took forward the registration process and was legally registered on 7th February 2022 at Helambu Rural Municipality.



नेपाल विद्युत प्राधिकरण
हेलम्बु मेलम्ची वितरण केन्द्र
दर्ता नं.: ५५१
मिति: २०७८।१२।०८

मिती

श्री कार्यलय प्रमुख ज्यू,
नेपाल विधुत प्राधिकरण
मेलम्चि शाखा ।

बिषय : नेट मिटरिङ प्रकृया अगाडि बढाउनको लागि सहजिकरण गरिदिनुहुने बारे ।

महोदय,

प्रस्तुत बिषयमा सोलार पीभी मिनिग्रिड प्रयोगकर्ताको समुह हेलम्बु गाउँपालिका धापसुङमा "धापसुङ महिला समुह" नामले दर्ता भएको छ । संगठनको दर्ता र नियमन ऐन २०२१ ले १५.७५ किलोवाट सौर्य पीवी डीसी-कपल्ड ब्याट्रीमा आधारित अफ-ग्रिड प्रणालीको हेरचाह गरिरहेको छ ।

अहिले राष्ट्रिय ग्रिड गाउँसम्म विस्तार भइसकेको छ तसर्थ धापसुङ महिला समुहले विद्यमान सोलार पीभी प्रणालीलाइ राष्ट्रिय ग्रिडमा जडान गर्न स्वीकृतिको लागि अनुरोध गर्न चाहान्छ ।

दिगो विकास इन्स्टिच्युट धापसुङ महिला समुहसँग नजिकबाट बिगत ५ बर्ष देखी काम गर्दै आएको छ र विधुत प्राधिकरणको मानक र प्राविधिक आवश्यकताहरु पुरा गर्ने १५k VA/3 फेज आउटपुट क्षमताको ग्रिड जोडिएको इन्भर्टरहरु प्रतिस्थापन गर्न जर्मनी दुतावास काठमाडौंबाट सहयोग प्राप्त गरेको छ ।

प्राधिकरणको निर्देशन २०७६ बमोजिमको प्रविधिक आवश्यकता प्रणालीले पुरा गर्नेछ तसर्थ हामी काम अगाडि बढाउन अनुमातिको आगि अनुरोध गर्न चाहान्छौ र मेलम्ची प्रधिकरणसँग समन्वयन गरी द्वि- द्विशात्मक मिटरिङ प्रउाली र आवश्यक कागजी कार्यहरुका लागि अनुरोध गर्दछौ ।

निवेदक

गीता
अध्यक्ष, धापसुङ महिला समुह
धापसुङ, हेलम्बु गाउंपालिका

Figure SEQ Figure * ARABIC 2: The letter submitted by Dhapsung women group to NEA Melamchi requesting to facilitate the net metering process

Two meetings had been arranged with Nepal Electricity Authority, the Melamchi distribution center to coordinate for the net metering process and the technical assessment of solar microgrid.

A field trip to Dhapsung village of Helambu rural municipality, Sindhupalchowk was organized on 25th May 2022. A team of 6 people including DBI team, Wind Power Nepal and a technician from Melamchi NEA (Nepal Electricity Authority) visited Dhapsung. The aim of this visit was to organize a technical assessment for prospective of Net metering of Solar micro grid in Dhapsung. The assessment stated that te two components of the grid connection a) Net Meter (3 phase) has to be scoped by NEA, Melamchi distribution and b) work on Automatic Transfer Switch (ATS) board for switching from NEA-Solar needs to be completed.

After the thorough inspection of the micro grid and conversation with NEA Melamchi official, technical experts and Dhapsung community, it was notified that besides the technical issues like a need to change the batteries and invertor, there are some policy loopholes on the net metering process. Net metering is the facility given to the costumer of NEA but in the case of Dhapsung, the net metering is proposed for community solar grid which is not a costumer of NEA. Additionally, a circular from the Nepal Electricity Authority dated July 1, 2018 stated that no net payment would be made to consumers if more energy were fed into the grid than consumed.

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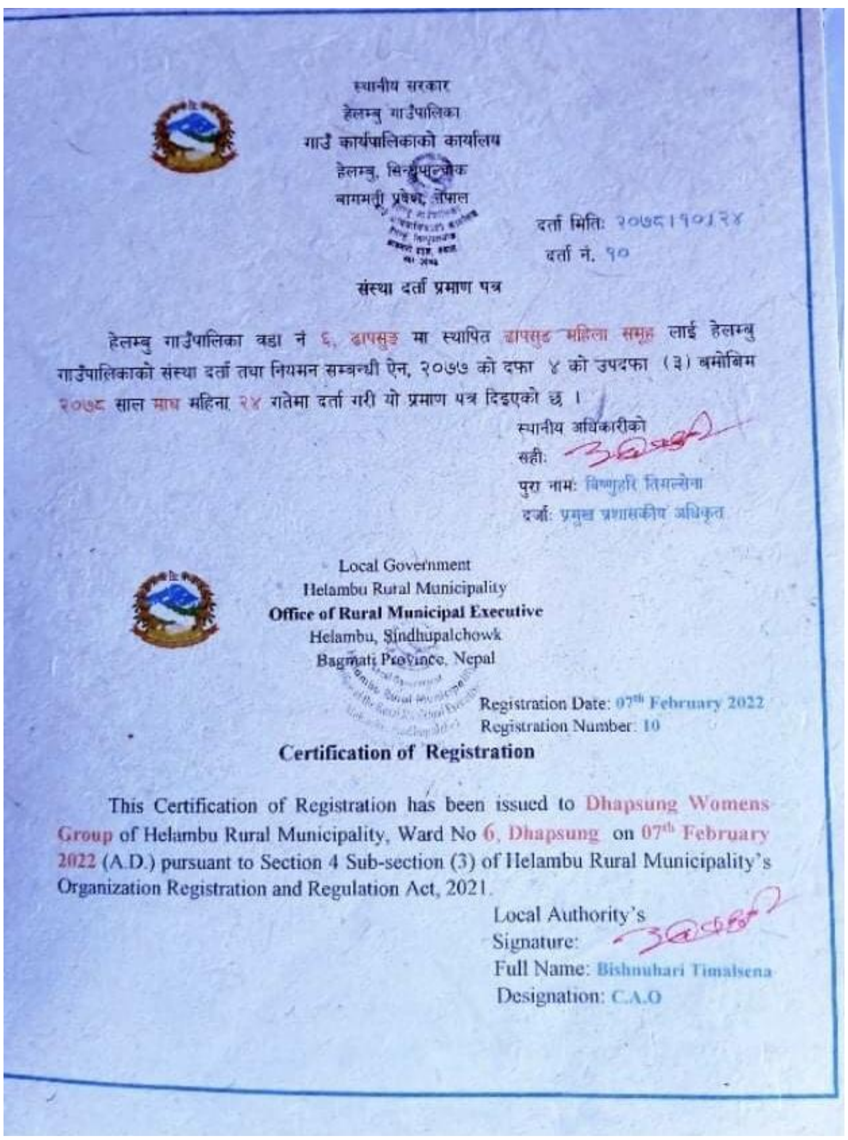

स्थानीय सरकार
हेलम्बु गाउँपालिका
गाउँ कार्यपालिकाको कार्यालय
हेलम्बु, सिन्धुपाल्चोक
बागमती प्रदेश, नेपाल

दर्ता मिति: २०७८।१०।२४
दर्ता नं. १०

संस्था दर्ता प्रमाण पत्र

हेलम्बु गाउँपालिका वडा नं ६, ढापसुङ मा स्थापित ढापसुङ महिला समूह लाई हेलम्बु गाउँपालिकाको संस्था दर्ता तथा नियमन सम्बन्धी ऐन, २०७७ को दफा ४ को उपदफा (३) बमोजिम २०७८ साल माघ महिना २४ गतेमा दर्ता गरी यो प्रमाण पत्र दिइएको छ ।

स्थानीय अधिकारीको
सही:
पुरा नाम: विष्णुहरि तिमल्सेना
दर्जा: प्रमुख प्रशासकीय अधिकृत

Local Government
Helambu Rural Municipality
Office of Rural Municipal Executive
Helambu, Sindhupalchowk
Bagmati Province, Nepal

Registration Date: 07th February 2022
Registration Number: 10

Certification of Registration

This Certification of Registration has been issued to Dhapsung Womens Group of Helambu Rural Municipality, Ward No 6, Dhapsung on 07th February 2022 (A.D.) pursuant to Section 4 Sub-section (3) of Helambu Rural Municipality's Organization Registration and Regulation Act, 2021.

Local Authority's
Signature:
Full Name: Bishnuhari Timalsena
Designation: C.A.O

Form

Figure SEQ Figure * ARABIC 3: Registration Certificate of Dhapsung women group for the net metering process

The technical experts suggested that DBI should organize a discussion program among the relevant experts and stakeholders to come up with the solutions to above mentioned policy loopholes. A panel discussion on "Challeneges of Net Metering and Way Forward" was organized on 19th Septmeber, 2022. During the Workshop the information on suspension of Net Metering facility was shared among us which has become the major obstacle in going forward with the Net metering process. Without any prior notification to the public, the NEA (Nepal Electricity Authority) abruptly discontinued net metering in July of 2022.

Figure SEQ Figure * ARABIC 4: Technical assessment and monitoring of Dhapsung solar micro grid

Panel discussion on "Challenges of Net Metering in Nepal and Way Forward"

The panel discussion on the "Challenges of Net Metering in Nepal and Way Forward" began with an introduction to the Solar Micro Grid installation in Dhapsung and the current challenges prevailing regarding the Dhapsun Community Solar Micro Grid. DBI in collaboration with wind power Nepal has been facilitating the connection of the national transmission line through net metering, to maintain the ownership rights held by the Dhapsung Women Committee in the energy system and involvement in the decision making process. In the context of Nepal, from the perspective of climate change, 5-10% of energy is utilized from the mini and micro hydro projects as per mentioned in NDCs (Nationally Determined Contributions).

A brief introduction by Ms. Gita Tamang, Supervisor of Dhapsung Women Committee, addressed about the before and after situation of the solar grid installed. She stated that the solar grid has not been in use for a few months due to the lightning events.

The panel discussion was forwarded by Mr. Kushal Gurung as the moderator. He requested the panelists to give their brief introduction and share their insights and experiences on the subject of Net Metering

Figure SEQ Figure * ARABIC 5: Panel discussion on Net metering challenges and way forward

Firstly, Mr. Jeevan Vaidya, Electronic and Communication Engineer, working in the sectors of Solar Water Pump and Rooftop Net Metering System, shared his journey up to the Green Interactive System and his experiences with the challenges and opportunities in various aspects of Net Metering System. He then mentioned about the installation of 1200 Solar Consumable systems in 63 districts of Nepal by his initiative cum private sector, Bridge Solar, since 2016. He mentioned that only 30% of energy is utilized from the installation of the solar water pumping system. In the context of Nepal, water pumping systems are used for irrigation purposes for only approx. 100 days, thus, the system remains idle in the remaining days rendering the waste of energy. Therefore, he expressed about the challenges to connecting with net metering and also, acknowledged the steps taken by DBI for Net metering in Dhapsung.

Next, Mr. Dilli Ghimire from Nepal Energy Foundation, briefly mentioned the joint efforts and initiatives carried out from his side for the micro hydro grid synchronization into net metering. Thereby, he stated about the two parts of Decentralization:

a) Technological decentralization- decentralized generation and distribution generation.

b) Distributed Decentralization

While working from the community side, he observed a few things: first while searching for ways to conserve distributed decentralization and make it sustainable, it was observed that that 37 MW generational capacity of mini-micro hydro projects. Secondly, there is a National Commitment and targets set in NDC to reach the aim of 100% renewable energy. He further mentioned that the issues of involving the solar and other mini- micro hydro to the national grid extension and assurance of its sustainability should be addressed more.

He underlined about the system of PPA. For this, mega documentation is required that consists of information such as the schedules, technicalities, and an application that ensures to meet the terms and conditions.

Figure SEQ Figure * ARABIC 6: Panel discussion on Net metering challenges and way forward

After that, Mr. Jeevan Kumar underlined the Electricity Act 2059 which stated that electricity of less than 100W cannot be connected to the national grid. The subject of the solar grid was raised after the 2016 A.D., therefore, a committee of professionals from IOE, NAST, and NEA was formed. After several meetings, the conclusion that the solar grid can be connected to the grid and run as a mega project was obtained, also, a 15% limitation was set as a thumb rule. Intermittent sources such as solar and wind result in energy fluctuation, thus, if a 15% limitation to inject into the grid, it can prevent energy fluctuation. As per the latest NEA report, with further works on solar grid interconnection at a mega scale, currently, 55 MW of electricity is injected into the grid. He further pointed out a few differences between the PPA and Net Metering systems. There are chances of price fluctuation in Net Metering compared to PPA. In Net metering, if the retail tariff changes, the export, and import rate vary accordingly. Talking about the process, Net metering is considered much easier than that of PPA.

The session was forwarded with a question and answer session involving the exchange of questions, answers and views between the panelists and the participants.

First question: Taking the example of the investment for integrating a 16 KW system into grid, is the return justified from the financial perspective by the investment? How sustainable is the investment from the dimension of return? Even after meeting all the conditions, how capable is a community after taking ownership of the system for its operation as it involves significant factors such as technical expertise, manpower, and operation cost?

Second question: Why has the issue of load shedding increased after grid synchronization at Tarakhola?

Addressing the first question, Mr. Jeevan Malik highlighted two terms: net metering and net billing and also briefly pointed out the differences between PPA and Net metering. In the PPA system, there is a provision for export only. For instance, if there is extra production of light, it cannot be sold to the community, whereas, in Net metering, the extra light produced can be sold to the community. Moreover, there was a system of 90% cap (for instance, if 100 unit is imported, the maximum unit that can be sold is only 90 unit) in net metering which has been removed.

While mentioning about the investment and return, he gave an example of a 16KW solar or micro mini hydro grid at a small scale and approximated the investment of 60 - 70 Lakh for it. Further the estimated calculation for return per day, month, and year, he stated that the investment will not be feasible when considered from an economical view.

Speaking of the second question, due to the extensive distance of the sub-station and long lines in Tarakhola, there are high possibilities of an electricity connection tripping whether due to voltage or other reasons which cause disconnection of the system.

Thereafter, Mr. Dilli Ghimire spoke about the commercial feasibility of net metering, in context of Nepal, he addressed that the issue is proceeding with a distributary approach. For instance, plenty of micro mini hydro projects and solar grid projects provide services to rural areas.

Referring to the contradictory management of "Bidhut Ain 2059 / Electricity Act 2059", he said that there is a separate act dedicated to Electricity Authority and it only follows the licensing rules mentioned in the Electricity Act.

Mr. Jeevan Vaidya introduced about the Feed-in Tariff and provided an instance of two metering system (one meter for measuring the electricity consumed and another to measure the electricity sold) which is a type of subsidy for the people in Oregon, Portland. Addressing the first question, he emphasized that even if at least 2 people are employed in the Operation and Maintenance sector, it could be considered an investment return.

Question by Mr. Kushal Gurung: He pointed out the fact that only 30% of electricity from the solar water pumping system is used for pumping. How the unused electricity can be related to Net metering in its present condition and utilized it? What recommendation can be given to the government of Nepal regarding it?

Mr. Jeevan Vaidya stated about the two dimensions of Solar Water Pumping: for the drinking water system and the irrigation system. Therefore, emphasized the point that Net metering is the only solution through which the aforementioned issue of unused electricity and the farmers can benefit significantly.

The question was directed to Mr. Dilli Raj Ghimire. What are the factors other than technical and economic factors that need to be considered while connecting micro hydro projects to the grid net metering? What can be done by the nation to prevent the wastage of the investment and idling of these projects after grid extension?

Mr. Dilli Raj Ghimire emphasized skill and knowledge transfer as major component that needs to be addressed more. He also pointed out the widespread system of providing only 3-4 days training to the people responsible to access a system. Next, he mentioned about the need of addressing "Distributor generation" in Nepal. If the possibilities of Distributor generation could be harnessed then, it will contribute for the Hydro based distributor generation, Biomass generation and Solar generation.

Questions directed to Mr. Jeevan Malik.

1. How can the approach of net metering be addressed from the perspective of fairness?
2. How can the legal loopholes in the net metering system be addressed?
3. How can the issues related to skill transfer and capacity building be addressed?

Figure SEQ Figure * ARABIC 7: Panelists expressing their views on Net metering in Nepal

Mr. Jeevan Malik highlighted and advocated about fairness in the system of net metering. If the dynamics of loss are fundamentally considered, import rates should be substantially higher than export rates and for this financial analysis is required. Talking about the issue of skill transfer, he mentioned that there is a lack of post-installation support (PIS) and because of this the community would not know where to go in case of damage to the system. He mentioned the ongoing efforts of NEA in mini-hydro projects to improve the institutional capacity, and skill enhancement in coordination with the rural municipalityFor the last session, the panelists formulated the way forwards.

Way Forwards by Mr. Jeevan Vaidya:

1. By adapting other dimensions and systems and setting clear guidelines, the solar grid system which is idle and not in use should proceed.
2. A customer committee could be formed by the Dhapsung Committee and registered in the ward, complying with the regulations in Dhapsung.
3. After that, an application for a 400V 3-phase meter (present- single phase meter) should be proposed.
4. If possible, the obligation for "Krishi Meter" should be proposed.
5. After that, an obligation for a Bidirectional meter should be given
6. Therefore, if the above-mentioned steps are accomplished, then, the Solar Mini Grid in Dhapsung can proceed sustainably.

Way Forwards by Mr. Dilli Ghimire:

1. Firstly, a business model should be designed and the demarcation of the energy services whether to use for cooking, lightning, or to utilize in the enterprises should be done by the Dhapsung women's committee along with a proper discussion with the local people.
2. The decision of whether the energy business at the community level is planned for the long term or not should be decided.
3. Also, the awareness regarding the primary concept and approach of net metering and PPA should be spread amongst the locals which would assist them in decision-making.
4. The decision of whether to go with the net metering or PPA, selling extra electricity to NEA, or buying electricity if not enough. Awareness should be spread amongst the local communities regarding the type and operation of the system.
5. If there is any potential customer who consumes 100 KW of electricity in the local area, then a rate negotiation should be made on handing over the operation authority of electricity and getting paid according to the rate fixed.

Way Forwards by Mr. Jeevan Malik:

He firstly highlighted about the dimensions through which the World Energy Trilemma Index views energy democracy:

- Energy security
- Energy equity and
- Environment sustainability

He emphasized the vulnerability of Nepal regarding electricity security and its major dependency on imported energy. The poor status of energy security is not only due to the dependence on imported energy but also, due to the lack of **Power generation mix**. He expressed about the need for debate at high levels regarding the current situation of energy security in Nepal.

Further, he highlighted about an analysis of the multi-tier framework provided by the World Bank and they are listed in points as follows:

- If tier 3 is considered as the baseline, there are few criteria that should be met such as should be connection of almost 200 watts, should be available for 8 hrs, etc.
- This analysis by world bank stated that the micro-mini hydro projects and solar grid projects lies above tier 3.

Referring to the way forwards, he recommended that since the net metering facility is halted by NEA, we should be focusing more on policy advocacy and pressurizing NEA to resume the net metering service.

If the net metering service is permitted by NEA again the further steps would be the submission of required documents such as Legally registered documents and certificates, community minutes, company recommendations and bank loan transactions. Then, a feasibility study should be conducted with voluntary site visit at Dhapsung further the submission of all of the abovementioned documents. The session was concluded with the closing remarks from Mr. Abhishek Shrestha, Program Director at Digo Bikas Institute.